



Final Project

Paper Reconstruction and Improvement: Time-Series Data

Course: Time-Series Analysis | Instructor: Havana Rika

Project Goal

In this part of the final project, each group will select a research paper related to time-series analysis, forecasting, classification, anomaly detection, change-point detection, time-series representation learning, etc. The goal is to reconstruct the method proposed in the paper, evaluate it correctly on time-series data, and then introduce a meaningful improvement.

The selected paper must include:

- A clear model, algorithm, or architecture that can be implemented.
- A public dataset, or a dataset that can be accessed and documented clearly.
- Experimental results that can be compared to your reconstruction.
- Preferably, an official GitHub repository or enough implementation details to reproduce the method.

Stage 1 - Reconstruct the Paper

Implement the method described in the paper and try to reproduce its main experimental results. The reconstruction must respect the temporal structure of the data: future observations must not be used during training, preprocessing, feature construction, or hyperparameter tuning.

Your reconstruction should include:

- A short explanation of the original method, including the model architecture, algorithmic steps, loss function, training objective, and key hyperparameters.
- A reproducible data pipeline: loading, cleaning, datetime parsing, resampling, missing-value handling, scaling/transformation, and feature construction where relevant.
- A valid temporal evaluation protocol, such as chronological train/validation/test split, rolling-origin evaluation, or walk-forward validation.
- At least one simple baseline, such as naive forecast, seasonal naive forecast, moving average, or a standard classical model when appropriate.
- Evaluation using the paper's metric, with an explanation of what the metric measures.
- Evaluation using at least one metric studied in class or appropriate for the task.

You do not have to match the exact numbers reported in the paper, but you must provide a serious comparison and explain possible differences, such as limited compute, different preprocessing, shorter training, different random seeds, unavailable hyperparameters, or dataset-version differences.

Stage 2 - Improve the Method

Modify the original method in a meaningful way and justify why the change should improve performance, robustness, interpretability, efficiency, or applicability to the time-series task. The improved method must be evaluated using the same dataset split and the same metrics as the reconstruction.

Examples of possible improvements:



- Improve preprocessing and data quality.
- Add stronger time-series features.
- Improve or tune the forecasting model.
- Improve ML or deep-learning architecture.
- Improve anomaly or change-point detection.
- Improve probabilistic forecasting and uncertainty estimation.
- Improve computational efficiency.
- etc.

Required Submission Files

1. Code for the reconstructed method, including data preprocessing, training, evaluation, and result reproduction.
2. Code for the improved method, including the changed architecture/algorithm and the same evaluation protocol.
3. The dataset, or a clear download link and instructions if the dataset is too large to submit.
4. A PDF report that explains the paper, the reconstruction, the improvement, and the results.
5. A short README file explaining how to run the code, including environment requirements, package versions, and expected outputs.

Important implementation requirements:

- All random seeds should be fixed where possible.
- Report the forecast horizon, sampling frequency, input window length, and output window length where relevant, and all additional hyperparameters.
- Clearly state whether the task is univariate, multivariate, supervised, or unsupervised, etc.

PDF Report Structure:

1. Original architecture: describe the model, loss, training objective, and important hyperparameters.
2. Paper results: summarize the reported results and metrics.
3. Reconstruction results: show your results and compare them to the paper.
4. Improved architecture: explain what you changed and why.
5. Improved results: compare (using a table) paper results, your reconstruction, and your improved model.
6. Discussion: explain what worked, what did not, and what you learned.
7. References.